SYSTEM REQUIREMENTS SPECIFICATION

Updated Functional & Non-Functional Requirements Document

PART I: REVISION LOG & SUMMARY OF CHANGES

1. What Was Added and Why

- **Authentication & Security:** Section 1 originally stated that users 'can log in' but never specified authentication mechanisms, security protocols, or role-based authorization rules. Security and access control are critical foundational requirements.
- **Admin Registration Period Management:** Section 11 mentioned enforcing registration only during valid periods, but lacked administrative capabilities to configure start/end dates. Admins require explicit control over these windows.
- **Waitlist Management:** When a course section reaches full capacity, blocking students outright creates operational friction. Real-world academic systems require an automated waitlist queue system.
- **Grade Appeal / Correction Workflow:** Section 19 allowed teachers to upload results, but provided no path to handle erroneous entries, disputes, or official grade changes post-publication.
- **Course Withdrawal Deadline:** Course dropping (Section 4) lacked time boundaries. The system must enforce academic calendar cutoff dates for course withdrawal.

2. What Was Modified and Why

- **Campus Smart Wallet Integration:** The existing Fee & Payment section (Section 16) was expanded into a unified Campus Smart Wallet system to handle daily micro-transactions across campus facilities. This transforms basic tuition viewing into a comprehensive campus payment ecosystem.

  - **Modified Wallet Structure Overview:**
  - Pay for food (canteen) -> Core use case — tap card/app
  - Pay for campus bus -> Core use case — fare from wallet
  - Pay for electricity (dorm) -> Core use case — top-up from wallet Wallet balance check -> See remaining money
  - Pay for electricity (dorm) -> Core use case — top-up from wallet Wallet balance check -> See remaining money
  - Transaction history -> Track spend and top-up
    Recharge / Top-up -> Add money via payment gateway

3. What Was Removed and Why

- **Section 18 Attendance Tracking:** Attendance tracking requires daily class-level logging, session management, and teacher inputs that belong within a Learning Management System (LMS) domain. Removing it prevents scope creep in the campus registration platform.
- **Section 22 Health & Support Services:** Health services (medical records, counseling appointments) involve sensitive medical privacy compliance and specialized healthcare workflows. It was removed to keep the document focused tightly on core campus administration.

PART II: FINAL REQUIREMENTS SPECIFICATION

1. User Roles, Permissions & Authentication

- **Student:** Can register, manage account, view academics, and perform wallet operations.
- **Teacher:** Can manage assigned courses, view rosters, and submit grades.
- **Advisor:** Can review, approve, reject student course registrations, and provide guidance.
- **Administrator:** Can manage users, courses, system settings, and registration periods.
- **Authentication & Security:** System enforces secure multi-factor user login, session management, password reset workflows, and Role-Based Access Control (RBAC).

2. Student Requirements

- **Account Management:** Student can create an account and update personal information.
- **Academic Records:** Student can view academic history, GPA, and degree progress.
- **Course Registration:** Student can search, register, drop courses, and join waitlists.
- **Notifications:** Student receives automated system alerts and announcements.

3. Course Management Requirements

- **Course Creation:** Admin can add new courses and delete obsolete courses.
- **Course Maintenance:** Admin can update course descriptions, credits, and rules.
- **Teacher Assignment:** Teacher can view assigned courses and class details.
- **Capacity Management:** System maintains course seat limits and prerequisites.

4. Course Registration Requirements

- **Search & Selection:** Student can search available courses and select sections.

- **Course Drop & Withdrawal:** Student can drop courses prior to the defined Course Withdrawal Deadline.
- **Prerequisite Checking:** System automatically checks and enforces prerequisites.
- **Seat Availability & Waitlist:** System checks seat availability and automatically queues students on a Waitlist when full.
- **Conflict Prevention:** System checks and prevents timetable schedule conflicts.

5. Recommendation Requirements

- **Academic Analysis:** System analyzes student academic history and completed courses.
- **Interest & Rules Matching:** System considers student interests, prerequisites, and degree requirements.
- **Recommendation Display:** Student can view personalized course recommendations.

6. Academic Advisor Requirements

- **Student Oversight:** Advisor can view assigned student academic information.
- **Registration Review:** Advisor can review, approve, or reject student registration requests.
- **Comments & Guidance:** Advisor can provide comments and course recommendations.

7. Teacher Requirements

- **Course Roster:** Teacher can view assigned courses and enrolled student lists.
- **Course Information:** Teacher can upload and update course syllabus/information.

8. Administrator Requirements

- **User & Course Admin:** Admin can manage students, teachers, courses, and system users.
- **Registration Period Management:** Admin can define, open, and close course registration windows.
- **System Rule Management:** Admin can manage global registration rules and limits.

9. Course Information Requirements

The system provides the following course attributes:

- **Fields:** Course Name, Course Code, Credit Hours, Teacher, Prerequisites, Course Description, Available Seats, Class Schedule, Classroom Location.

10. Student Information Requirements

The system maintains the following student profile fields:

- **Fields:** Student ID, Name, Department, Program, Completed Courses, Current Courses, Academic Results, Credits Completed, Remaining Credits.

11. Registration Rules

- **Prerequisite Validation:** Checks prerequisite completion.
- **Credit Limits:** Enforces maximum and minimum credit limits.
- **Capacity Control:** Enforces course capacity and waitlist management.
- **Conflict & Duplicate Prevention:** Detects timetable conflicts and prevents duplicate course registration.
- **Registration Window:** Allows registration exclusively during active registration periods set by Admin.

12. Timetable Requirements

- **Schedule Generation:** System generates registered-course timetables automatically.
- **Display Details:** Student can view weekly class schedule, times, and classroom locations.
- **Conflict Detection:** System detects schedule overlaps.

13. Notification Requirements

- **Triggers:** System notifies users about registration deadlines, approval/rejections, waitlist updates, course changes, timetable shifts, new recommendations, and academic announcements.

14. System Outputs

- **Outputs:** Registration Confirmations, Student Timetables, Course Recommendation Lists, Registration Summaries, Academic Summaries, Notifications, and System Reports.

15. Dormitory & Accommodation

- **Student Features:** Student can apply for dormitory rooms and view available rooms/status.
- **Admin Features:** Admin can manage room allocations and accommodation status.

16. Campus Smart Wallet & Payment Requirements

- **Pay for Food:** Student can tap card or app to pay at campus canteens/cafeterias.
- **Pay for Campus Bus:** Student can pay bus fares directly from wallet.
- **Pay for Electricity:** Student can top-up electricity for dorm rooms from wallet.
- **Wallet Balance Check:** Student can view real-time remaining wallet balance.
- **Transaction History:** Student can view full spend and top-up transaction logs.
- **Recharge / Top-up:** Student can add funds to wallet via online payment gateway.

17. Library Requirements

- **Catalog & Borrowing:** Student can search books, view borrowed books, and track due dates.
- **Reservations & Alerts:** Student can reserve books and receive overdue notifications.

18. Exam, Result & Grade Appeal Requirements

- **Exam Schedule:** Student can view published examination schedules.
- **Result Publication:** Teacher/Admin can upload academic results; student can view results.
- **GPA Calculation:** System automatically calculates GPA/CGPA and maintains academic history.
- **Grade Appeal / Correction Workflow:** System provides a formal process for students to submit grade appeals and for teachers/admins to issue approved grade corrections.

19. Campus Life & Events

- **Events:** Student can view and register for campus events; Admin can create/update events.

20. Communication & Collaboration

- **Messaging:** Students and advisors/teachers can communicate directly via messaging; Admin can issue system announcements.

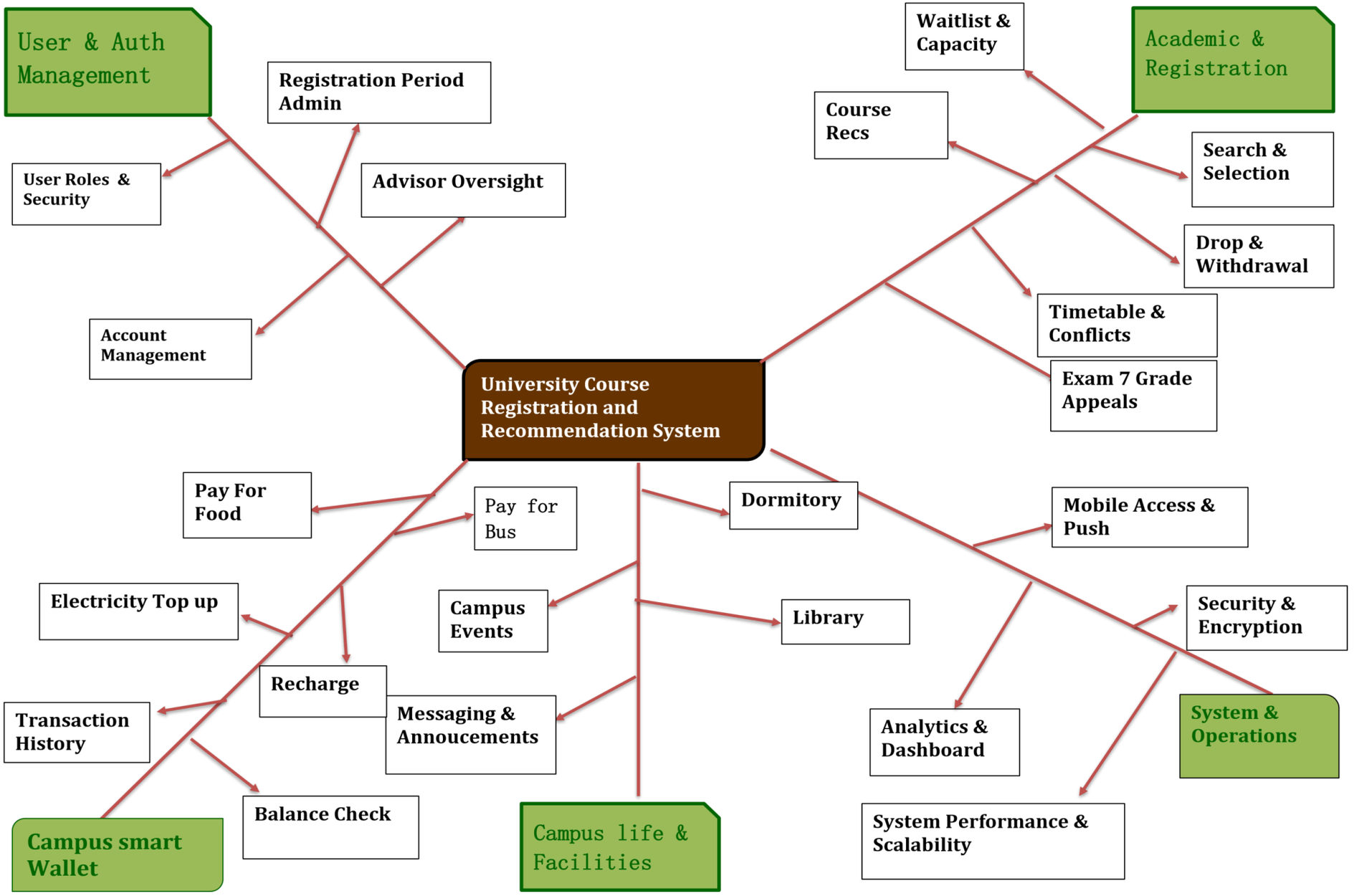
21. Mobile & Accessibility

- **Mobile Access:** Student can access registration and academic features via mobile devices.
- **Mobile Push:** System delivers mobile alerts and notifications.

22. Analytics & Personalization

- **Academic Insights:** System analyzes student performance and generates insights.
- **Dashboard:** Student can view personalized dashboard and track registration patterns.

❖ Feature Tree



PART III: NON-FUNCTIONAL REQUIREMENTS

- **Performance:** System must process registration requests rapidly and handle concurrent peak loads.
- **Security:** Data encryption in transit and at rest; secure session management and RBAC.
- **Reliability:** System must maintain high data consistency and zero data loss during transactions.
- **Availability:** Target system uptime of 99.9% during active registration windows.
- **Usability:** Intuitive user interface for both desktop and mobile platforms.
- **Scalability:** Infrastructure capable of expanding capacity with growing student intake.
- **Maintainability:** Modular architecture enabling updates without system downtime.
- **Accessibility:** Compliance with web accessibility standards for diverse user capabilities.